

ELEC 6910D Lab: Design Flow with Commercial EDA Tools

1 Overview

In this lab, you will be provided with a simplified default EDA flow to experience the basic implementation of digital design. It involves both logic synthesis and physical design (floor-planning, placement, clock tree synthesis, and routing). We provide you access to Synopsys Design Compiler for logic synthesis and Cadence Innovus for physical design. These tools are controlled with design flow scripts (filename in `***.tcl`) in *Tcl* programming language. In addition to the design flow and tools, you will be provided with three simple designs (i.e., TinyRocket, FPU, Sasc) in Verilog format, at the initial RTL stage. This lab tentatively accounts for around 20% of total course grade.

This lab consists of two tasks, as summarized below.

- Task 1 (basic, 60% grade): Finish the implementation of three given designs, directly using our provided simplified default EDA flow, which is already automated. **Report** the ultimate (i.e., post-routing) design quality, including timing, power, area (PPA), and DRC violations. Comment on the quality or problems of such chip implementation.
- Task 2 (advanced, 40% grade): Try to revise the provided default EDA flow (in `.tcl` files) and implement three designs with revised flow again. **Report** what you do and ultimate chip results. You may target achieving better implementation results (e.g., better PPA, less violations, etc.) or fixing existing problems in the default flow, if you notice some. Or you may just do some experiments to evaluate the impact of certain optimizations, and report your discoveries. Just play with it and have fun!

You are supposed to submit materials below. Please put them in a zip file.

1. A report in `.pdf` describing your implementation and results, minimum 2 pages, maximum 6 pages, no required template. Follow requirements in above 2 tasks. Describe the area, power, timing, and DRC of both tasks and make comments on them. Describe the changes/improvement you have made and why you tried it. Describe any other discoveries or your opinions.
2. The screenshot of each design layout after routing. Include as image files.
3. The report file of area, power, timing, and DRC. Include as plain text files.

For technical questions, please email our TAs and CC course instructor:

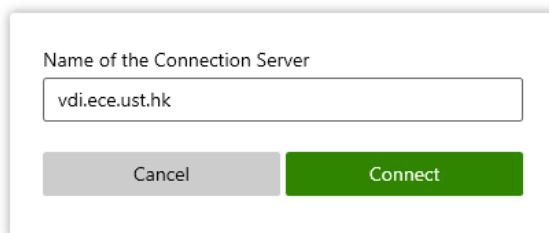
1. TA: Wenji FANG (`wfang838@connect.ust.hk`)
2. TA: Yao LU (`yludf@connect.ust.hk`)
3. Instructor: Zhiyao XIE (`eezhiyao@ust.hk`)

2 ECE LAB setup procedure

The installation guide of the VMware horizontal client is available in the link below:

<https://itsc.hkust.edu.hk/services/academic-teaching-support/facilities/virtual-barn/installation>

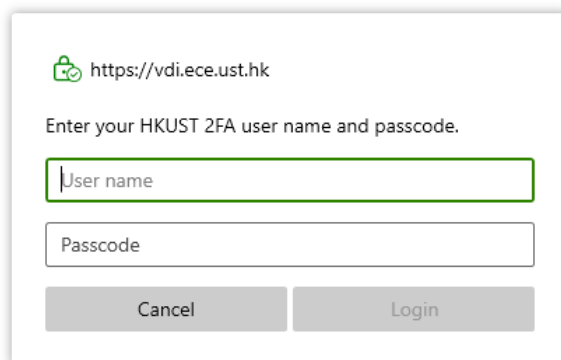
To connect to “ECE LAB7”, typing “vdi.ece.ust.hk” when adding the server.



Name of the Connection Server

Cancel Connect

Enter the account name, (not the whole email address). For example, if for ITSC account `eeabc@ust.hk`, just input “`eeabc`”.

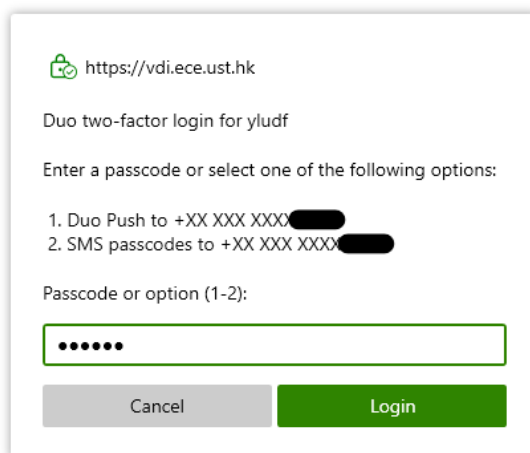


 https://vdi.ece.ust.hk

Enter your HKUST 2FA user name and passcode.

Cancel Login

Afterward, a DUO 2FA check-up is prompted.



 https://vdi.ece.ust.hk

Duo two-factor login for ylundf

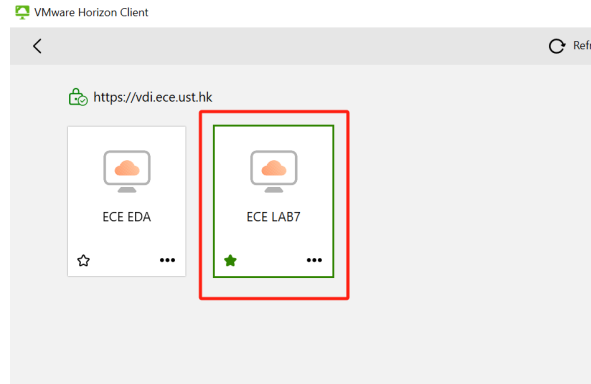
Enter a passcode or select one of the following options:

1. Duo Push to +XX XXX XXXX 
2. SMS passcodes to +XX XXX XXXX 

Passcode or option (1-2):

Cancel Login

When successful, select “ECE LAB7” pool.



3 Run EDA flow

3.1 Build the EDA environment

First, please copy the lab file from the course public folder and unzip it to your own folder.

```
$ cp /staff/ee/dept/public/elec6910d/ELEC6910D_Lab.zip /your_folder
$ cd your_folder
$ unzip ELEC6910D_Lab.zip
```

The required EDA tools are available in the ECE LAB VDI environment.

1. Synopsys Design Compiler

```
source /usr/eelocal/synopsys/syn-vv2023.12/.cshrc
```

2. Synopsys PrimeTime

```
source /usr/eelocal/synopsys/pts-vt2022.03-sp5/.cshrc
```

3. Cadence Innovus

```
source /usr/eelocal/cadence/innovus201/.cshrc
```

We have also provided a friendly and convenient method. Simply run **'source setup.sh'** provided in our folder ELEC6910D/.

Tips: If you are not familiar with any given command for Design Compiler, you can use the command **"help get*"** after entering *dc_shell* prompt environment to perform a fuzzy search and match commands. Additionally, you can use **"man get_cells"** in the prompt environment to obtain detailed information about "get_cells". The same applies to Innovus (*innovus*). Also, you can find 'help' option in the GUI.

```

dc_shell> man get_cells
2. Synopsys Commands
get_cells
Command Reference

NAME
    get_cells
    Creates a collection of cells from the current design, relative
    to the current instance.

SYNTAX
    collection get_cells
        [-quiet]
        [-regexp | -exact]
        [-nocase]
        [-filter expression]
        [-hierarchical]
        [-rtl]
        [patterns
        | -of_objects objects
        | -within region
        | -touching region

```

3.2 Detailed Information of each folder

After setting the environments, you can see two folders: **Flow/** and **rtl/**.

In rtl/, we provide you with some RTL designs. Or you can put your own designs in this folder.

In Flow/, there are three subfolders: **dc_release/**, **innovus_release/**, and **tech/**.

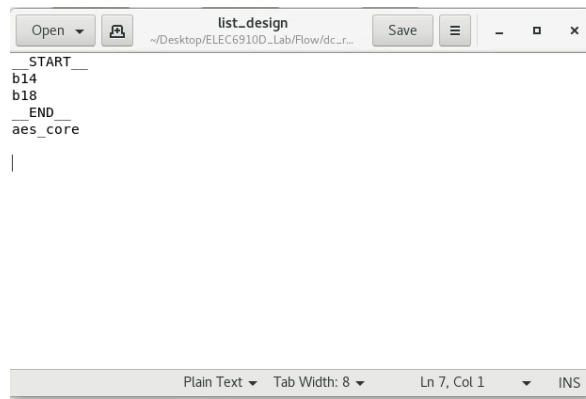
- dc_release/: Design Compiler flow.
- innovus_release/: Place and Routing flow
- tech/: stores the library files (no need to change). We adopt open-source tech library **Nangate 45nm** in this lab. FYI, our ECE IC Design Center (ICDC) provides TSMC technology libraries in 22nm, 28nm, 40nm, 65nm, etc. But you need to sign NDA before accessing it.

3.3 Run the flow

3.3.1 Logic synthesis

```
$ cd Flow/dc_release
```

In the **'list_design'** file, you can put the design name you need to run between **'_START_'** and **'_END_'**.



Parameter adjustment is done in the ‘**dc.tcl**’ file. You can try setting different parameters to constrain various aspects of the circuit, such as the clock or area, among others. This will have an impact on your synthesis results. Feel free to experiment and explore!

To ensure that you are in the Bash environment, you can use the command “echo \$0” to check. If you are not in Bash, please enter “bash” to switch to it. Make sure the following commands are executed successfully.

```
$ source run_dc.sh
```

This is a one-click run command. If there are no errors, you should get your DC results in a few minutes.

3.3.2 Placement and Routing

Similar to logic synthesis flow, you can run the P&R flow by the following steps.

```
$ cd Flow/innovus_release
```

In the ‘**list_design**’ file, you can put the design name you need to run between ‘_START_’ and ‘_END_’.

Parameter adjustment is done in the ‘**pnr.tcl**’ file.

Tips: We recommend running **one design** at a time to facilitate viewing the visual results. If you want to run all three designs at once, please remove the commented “exit” statement at the end of the *pnr.tcl* script.

```
$ source run.sh
```

This is also a one-click run command. If there are no errors, you should get your P&R results.

3.4 Check Results

You will find a folder with the same name as the design in the current directory after the completion of synthesis(DC and P&R).

In this folder, you can explore the “reports” and “results” directories to view outputs, such as netlists, area, power, and timing reports.

